22	**D** Electric force $F_E = qE = q(\Delta V/d)$ on the oil drop due to the electric field is upwards. When p.d. is V_1 and oil drop is moving up at constant speed, qV_1/d – mg – resistive force = 0 ➔ resistive force = qV_1/d – mg ……………... (1) When p.d. is V2 and oil drop is moving down at the same constant speed, qV_2/d – mg + resistive force = 0 ➔ resistive force = mg – qV_2/d ……………… (2) Solving, we have $q = 2mgd / (V_1 + V_2)$.
23	**D**